

Jonathan Kang

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Palatine, Illinois

EDUCATION

University of Illinois at Chicago

December 2025

Bachelor of Science in Computer Science

GPA: 3.71/4.00

Relevant Coursework: Database Systems, Advanced Data Structures and Algorithms, Concurrent Computing, High Performance Computing, Computing Algorithms, Computer Graphics Programming, Calculus III, Object-Oriented Programming, Computer Design, Software Design

SKILLS

Certifications: AWS Cloud Practitioner

Programming Languages: Python, C, C++, Java, JavaScript, CSS, HTML, R

Tools: AWS, Docker, PostgreSQL, SQLite, Git, PowerShell, React, Postman

Languages: English, Korean

PROJECTS

Wine Query Answering System

October - December 2025

- Built a semantic question-answering application over wine reviews using a Python backend, TypeScript frontend, and a PostgreSQL database.
- Designed a vector embedding database pipeline to chunk wine reviews and enable nearest-neighbor vector retrieval.
- Implemented relevance ranking using cosine similarity and dot product, returning the top 5 most relevant passages per query.

Fish and Fitness

February - May 2025

- Produced a web application to create a goal-focused, fish-themed extension of the Strava exercise app, using Django/React+JavaScript and the SQLite database.
- Utilized the Strava API to extract user data for the web application.
- Applied a probabilistic data structure called Cuckoo Filter, speeding up the username searching process by up to 50% and reducing the need to search the username database.

Local Restaurant Discovery Site

February 2025

- Designed a web application to show local, lesser-known restaurants at the SparkHacks 2025 hackathon, using Django/React with TypeScript as the front end and Python as the back end.
- Validated data accuracy by retrieving statistics from the Google Search and Google Places APIs.
- Implemented restaurant discovery customization by adapting to specified labels and previous user choices to speed up browsing by 30%.

3-Card Poker

November - December 2024

- Collaborated with a partner to create 3-card poker using Java's JavaFX and FXML.
- Built with Apache Maven, the program allows players to run the game simultaneously on a hosting server, with concurrency accounted for.
- Constructed the calculation behind the betting system through ante, pair plus, and playing bets.
- Unit-tested code with 250+ tests using JUnit 5 to test the server's cards and betting logic.